

Email SOAR Automation

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1. Introduction

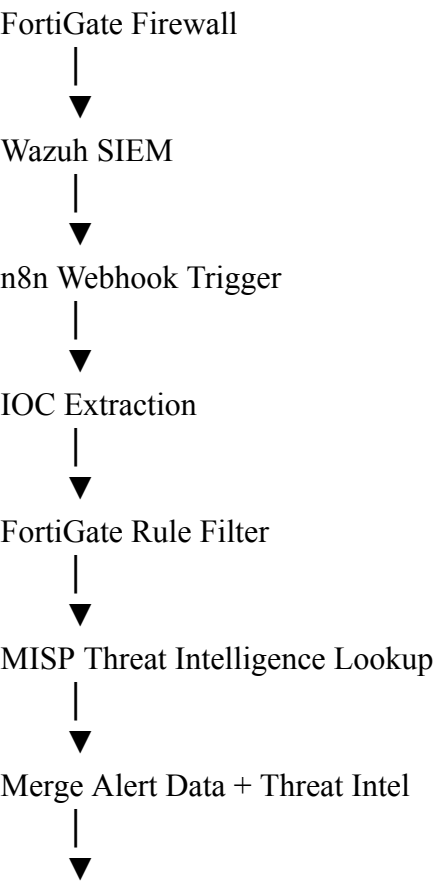
This project implements a **Security Orchestration, Automation and Response (SOAR)** workflow that integrates **Wazuh SIEM**, **MISP Threat Intelligence Platform**, and **automated email alerting** using the **n8n workflow automation engine**.

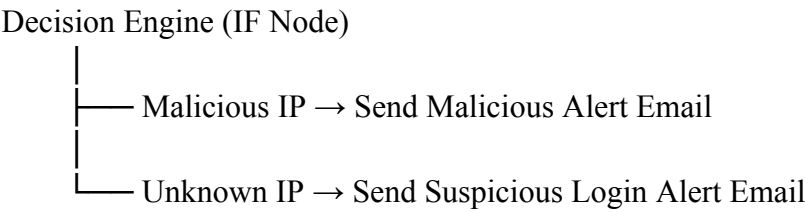
The automation processes security alerts generated by **FortiGate firewall logs** monitored through Wazuh. The workflow extracts **Indicators of Compromise (IOC)** such as the source IP address and verifies them against **MISP threat intelligence**. Based on the lookup results, the system automatically sends an email notification to the security analyst.

This solution improves **incident response speed**, reduces **manual analysis**, and enhances **security monitoring efficiency**.

2. System Architecture

The overall architecture of the automation pipeline is shown below:





3. Tools and Technologies

Component	Description
Wazuh	Open-source SIEM used for log monitoring and security analytics
n8n	Workflow automation platform used to implement SOAR logic
MISP	Malware Information Sharing Platform for threat intelligence
FortiGate	Firewall generating security event logs
Email (SMTP/Gmail)	Used for automated alert notifications

4. Workflow Implementation

The automation workflow is built using **n8n nodes**.

The following nodes are implemented in the workflow.

4.1 SIEM Webhook Node

The **Webhook node** acts as the entry point for the automation workflow.

It receives **security alerts from Wazuh** using HTTP POST requests.

Example alert payload:

```
{
  "timestamp": "2026-03-04T09:58:00",
  "rule": {
    "level": 8,
    "description": "FortiGate Authentication Failure"
  },
  "data": {
    "srcip": "192.168.0.62"
  }
}
```

Once the webhook receives the alert, the workflow execution begins.

4.2 Extract Source IP Node

This node extracts important security event information from the incoming JSON payload.

Extracted fields include:

- Source IP address (**srcip**)
- Rule description
- Severity level
- Timestamp

Example output:

```
srcip: 192.168.0.62
rule_desc: FortiGate Authentication Failure
severity: 8
timestamp: 2026-03-04
```

These fields are used in later stages of the workflow.

4.3 Filter Specific FortiGate Rule Node

This step filters incoming alerts so that only **relevant FortiGate events** are processed.

Example filtered events:

- Login failures
- Authentication errors
- Suspicious access attempts

This reduces false positives and unnecessary alert processing.

4.4 MISP Threat Intelligence Lookup

The extracted **source IP address** is queried against the **MISP threat intelligence database**.

API Endpoint

POST /attributes/restSearch

Example Request

```
{
  "type": "ip-src",
  "value": "192.168.0.62"
}
```

Example Response

```
{
  "response": []
}
```

An empty response indicates that the IP address is **not present in the threat intelligence database**.

4.5 Merge Node

The **Merge node** combines:

- Original alert data
- MISP lookup results

This ensures the workflow retains all necessary information for decision making.

Example merged output:

```
{
  "srcip": "192.168.0.62",
  "rule_desc": "FortiGate Authentication Failure",
  "severity": 8,
  "timestamp": "2026-03-04",
  "response": []
}
```

4.6 Decision Engine (IF Node)

The **IF node** evaluates whether the IP address exists in the MISP threat intelligence database.

Condition Used

`response.length > 0`

Logic

Condition

Result

TRUE	Malicious IP detected
FALSE	IP not found in threat intelligence

4.7 Automated Email Alerts

Based on the decision result, the workflow sends one of two types of alerts.

Malicious IP Alert

If the IP exists in MISP:

Malicious IP Detected

Source IP: 192.168.0.62
Threat Intel: Found in MISP
Severity: High

Recommended Action:
Investigate immediately and block the IP.

Suspicious Login Alert

If the IP is not found in threat intelligence:

Suspicious Login Observed — Not in Threat Intel

Source IP: 192.168.0.62
Rule: FortiGate Authentication Failure
Severity: Medium
Timestamp: 2026-03-04

Threat Intel: Not found in MISP

Recommendation:
Monitor repeated authentication failures.

5. Security Automation Benefits

This automation provides several operational advantages:

- Automated IOC analysis
 - Faster incident detection
 - Reduced manual SOC workload
 - Real-time threat intelligence correlation
 - Improved security monitoring efficiency
-

6. Limitations

The current implementation relies only on **MISP threat intelligence**. If the IP is not present in the database, the alert will be classified as suspicious but not malicious.

7. Future Enhancements

The following improvements can enhance the system:

Threat Intelligence Enrichment

Integrate additional sources such as:

- AbuseIPDB
- VirusTotal
- AlienVault OTX

GeoIP Lookup

Determine attacker geographic location.

Automated Firewall Blocking

Automatically block malicious IP addresses on FortiGate firewall.

SOC Dashboard Integration

Send alerts to platforms such as:

- Slack
 - SIEM dashboard
 - Incident management systems.
-

8. Conclusion

This project demonstrates how **SIEM alerts can be automated using SOAR techniques**.

By integrating **Wazuh, n8n, and MISP**, the system can automatically analyze firewall alerts, enrich them with threat intelligence, and generate actionable alerts for the security operations team.

The solution improves **incident response speed**, reduces **manual investigation effort**, and enhances **security monitoring capabilities**.

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